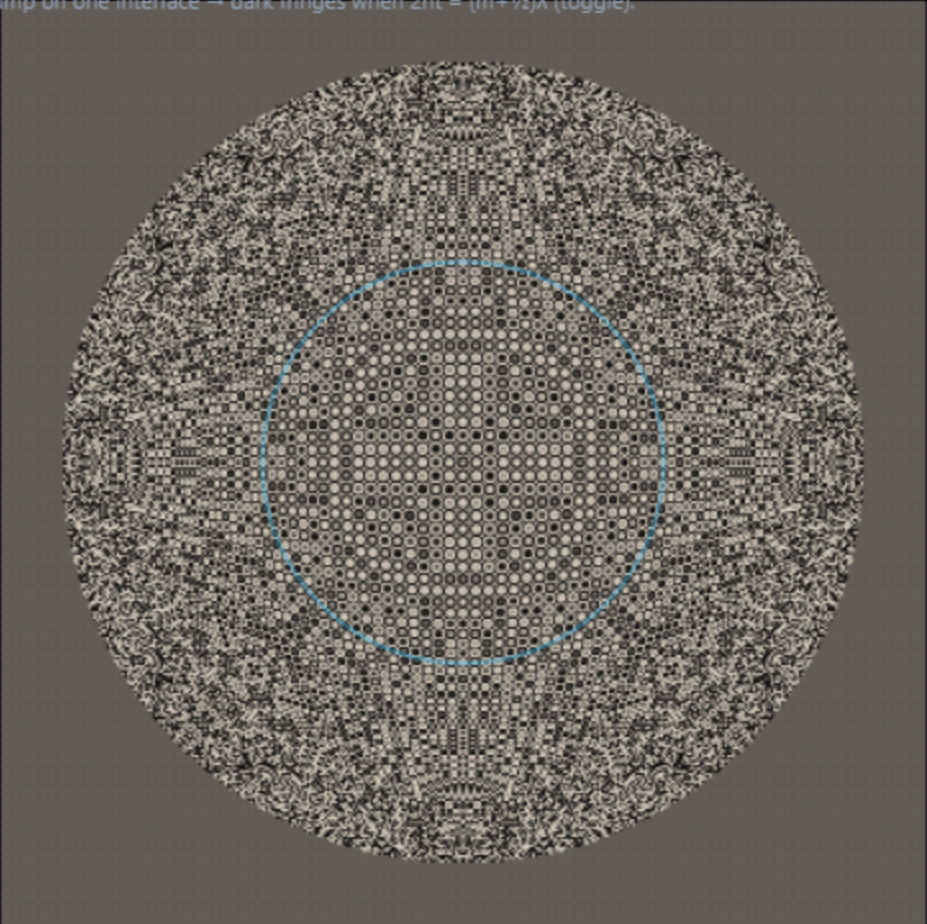


Newton's Rings

Spherical lens on flat glass: air-film thickness fringes; optional π phase on reflection.

Reflection from air film + glass: phase jump on one interface $\rightarrow$ dark fringes when $2nt = (m+\frac{1}{2})\lambda$ (toggle).



FILM & LENS

λ 540 nm

Lens radius R1.18 mm

Gap at center d_0 160 nm

Film n1.06

☒ π phase jump on reflection (glass side)

Paraxial spherical gap; qualitative reflectance $\cos^2(\delta/2)$. Real rings use extended-source geometry and finite coherence.

MEASURED VALUES

$\approx$ 1st dark ring r (small-angle)0.02 μm

About this model

Newton's rings appear when a slightly curved optical surface (often a plano-convex lens) contacts a flat plate, trapping a thin air film whose thickness grows with radius. Light reflected from the top and bottom of the film interferes. For near-normal viewing the optical path difference is about $2nt$, with t the local gap height and n the film index (air ≈ 1). A π phase shift on reflection from the lower-index side of an interface can invert bright/dark conditions; the checkbox models the common case of a phase jump when reflecting from the glass plate. The circular symmetry makes fringes concentric; the small-angle radius of the m -th dark ring scales like $\sqrt{(m\lambda R/n)}$ when the extra π is included. The raster

image uses $\cos(4\pi nt/\lambda + \text{optional } \pi)$. Assumptions: quasi-monochromatic light, scalar interference, no finite coherence or oblique incidence — adequate for qualitative lab demos.

Who it's for: High school and college optics labs studying thin-film interference, phase shifts on reflection, and radius-of-curvature measurements.

KEY TERMS

- Newton's rings
- Thin film
- Interference
- Phase shift on reflection
- Optical path difference
- Plano-convex lens
- Air wedge
- Fringe radius

How it works

Concentric interference fringes from a thin air wedge between a spherical lens and flat glass: optical path difference $2nt$ plus possible π phase on reflection.

Frequently asked questions

Why are the fringes circular?

The lens–plate gap thickness depends only on distance from the contact point (to good approximation), so loci of equal thickness are circles centered on the contact.

What does the π checkbox change?

If only one of the two reflections picks up a π phase (low-to-high index), bright and dark conditions swap. Toggling the box lets you match either textbook convention.

Can Newton's rings measure lens curvature?

Yes. Measuring ring diameters and knowing λ yields the radius of curvature R of the lens surface via the small-angle fringe spacing relation.

Why does the simulation look pixelated?

It uses a finite grid for speed. High-resolution cameras in the lab see smooth rings until diffraction limits intervene.

Related simulators

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NEW

SCHOOL

Thin-Film Interference

▶

Launch Simulator

Wedge fringes: n , d , θ ; $\cos^2(\delta/2)$ colors and $I(\lambda)$ at mid thickness.

NEW

SCHOOL

Anti-Reflection Coating

▶

Launch Simulator

Air–film–glass at normal incidence: $R(\lambda)$ from two-interface interference; quarter-wave design.

NEW

UNIVERSITY / RESEARCH

Michelson Interferometer

▶

Launch Simulator

$I(\Delta) = V \cos^2(\pi\Delta/\lambda)$; tilt fringes; coherence length envelope.

NEW

UNIVERSITY / RESEARCH

Mach–Zehnder Interferometer

▶

Launch Simulator

Two-beam recombination; $I \propto \cos^2(\pi\Delta/\lambda)$ vs one-arm OPD; schematic + fringe plot.

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Optical Bench (sandbox)

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Up to 4 elements: thin lenses, vertical mirrors, wedge δ ; paraxial ray trace.

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Reflection

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Flat, concave, and convex mirrors with auto-drawn ray diagrams.